Development Roadmap: Smart Agriculture Deep Learning System

Phase 1: Planning & System Architecture

Define what your system will do: input NPK/pH values, analyze leaf/soil images, consider location & weather, and return crop/fertilizer suggestions.

Tasks:

- Sketch user flow
- Break project into modules: Frontend, Backend, ML, API

Phase 2: Learn the Fundamentals

Learn frontend: React Native or MIT App Inventor

Learn backend: Python Flask + MongoDB

Learn ML basics: CNN, image classification

Learn geolocation and weather API usage

Phase 3: Start Building Each Component

Frontend:

- Build mobile interface with form inputs
- Use React Native/MIT App Inventor

Backend:

- Flask APIs to receive input, return suggestions
- Integrate MongoDB for storage

AI/ML:

- Train CNN model for leaf disease or deficiency detection
- Use TensorFlow/PyTorch

Geolocation:

- Use GPS + weather APIs (OpenWeather/NASA POWER)

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- Use Leaflet.js/GeoPandas for maps

Phase 4: Test & Iterate

Test image models with real photos

Try different NPK values

Check responses to different GPS/weather inputs

Collect user feedback from farmers or students

Phase 5: Finalize, Deploy & Prepare for Demos

Deploy backend on Render/Heroku

Generate APK for Android

Create pitch deck (problem, solution, tech stack, impact)

Get ready for demo or competition

Advanced Features (Future Scope)

Voice assistant (Google Speech API)

Drone monitoring

Blockchain traceability

Financial yield predictor using ML

Startup Grants & Funding Opportunities

Startup India Seed Fund - Rs.20-50L

DST NIDHI PRAYAS - Rs.10L

FICCI + Samsung - Mentorship + toolkit

NABARD - Rural pilot testing support

IIT Agri Incubators - Labs + early seed funding